

Portfolio

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Overview of Projects

Project 1: Designing Social Robots for Mental Health

Project 2: How AI Error Types Shape Trust, Reliance, and Compliance

Project 3: Behavioral and fMRI Analysis of Trust and Reciprocity

Additional Projects and Experience

Project 1: Designing Social Robots for Mental Health

Conducted a critical review of animal- and robot-assisted therapy research to identify how interaction design, engagement, and scalability shape therapeutic outcomes. Synthesized clinical findings through a human-factors lens to evaluate which design choices support—or limit—long-term user engagement.

Background

Current Landscape

- Loneliness and social isolation significantly worsen outcomes across multiple mental health conditions, creating demand for scalable interventions
- Animal-assisted therapy (AAT) shows benefits, but faces scalability, cost, and ethical constraints; social robots are increasingly proposed as alternatives

Knowledge Gap

- Clear comparisons between the effectiveness of robot-assisted and animal-assisted therapies across populations
- Understanding of which design and interaction features support sustained user engagement over time.

Future Relevance

- Guide when and how robot-assisted therapy should be deployed in mental healthcare settings
- Inform design priorities for social robots intended for emotionally sensitive use cases

Research Question: How effective are robot-assisted therapies compared to animal-assisted therapies, and what design and human-factors limitations currently constrain their impact?

Specs

Role:

- Primary author, responsible for defining the review scope, framework, and analytical approach
- Synthesized interdisciplinary evidence and translated findings into human-factors and design-relevant insights

Methods:

- Critical narrative literature review of peer-reviewed studies on animal- and robot-assisted therapy
- Comparative analysis across populations, intervention types, and robot design characteristics

Participants:

- Secondary research only; reviewed studies included children, adults, and older adults across psychiatric and neurodevelopmental populations

Timeline:

- Literature review, synthesis, and writing conducted over approximately 12 months

Challenge:

- Highly heterogeneous study designs and outcome measures limited direct comparison
- Many studies prioritized feasibility over long-term effectiveness or engagement

Key Insights & Takeaways

Insights:

- Across studies, animal-assisted therapy shows more consistent short-term benefits than robot-assisted therapy, while robot interventions remain promising but uneven in effectiveness and engagement
- The primary gap across robot-assisted therapy research is sustained user engagement, driven by limited adaptability, shallow interaction, and insufficient attention to human factors

UX Impact:

- Identifies limited emotional depth, repetitive interactions, and declining engagement over time as primary UX constraints on the long-term effectiveness of therapeutic robots
- Highlights the need for rigorous UX-focused studies—including controlled trials and longitudinal designs—to evaluate engagement, emotional responsiveness, and real-world use beyond feasibility
- Indicates that effective therapeutic robots must prioritize emotional responsiveness, personalization, and adaptive interaction over anthropomorphic appearance alone

Reflections:

- Emphasized the importance of evaluating user experience outcomes, not just feasibility or novelty, in early-stage human-AI systems
- Reinforced that human factors and interaction design are central to long-term engagement and meaningful impact in emotionally sensitive technologies

Project 2: How AI Error Types Shape Trust, Reliance, and Compliance

Designed and implemented a web-based human-AI interaction experiment to examine how different AI error types influence user trust, reliance, compliance, and correction behavior. Built the full study in jsPsych and JATOS, including task flow, AI feedback logic, knowledge checks, and post-task surveys.

Background

Current Landscape

- AI systems increasingly support human judgment in high-stakes and everyday decision-making contexts
- User trust in AI is shaped not only by accuracy, but by how and when systems make errors

Knowledge Gap

- Limited understanding of how different AI error types (e.g., false alarms vs. misses) differentially affect trust, reliance, compliance, and correction behavior
- Insufficient design guidance for calibrating trust when AI systems make unavoidable errors

Future Relevance

- Informs how AI systems should communicate uncertainty and recover from errors without undermining user confidence
- Supports UX research aimed at designing AI tools that promote appropriate reliance/compliance rather than blind trust or disengagement

Research Question: How do different AI error types influence user trust, reliance, and corrective behavior during human-AI interaction?

Specs

Role:

- Sole developer of the experimental platform; implemented task flow, AI advice logic, knowledge checks, and surveys
- Translated human-factors research questions into a fully functional jsPsych and JATOS experiment

Methods:

- Designed a between-subjects human-AI interaction experiment manipulating AI error type (false alarms vs. misses)
- Implemented interactive task trials, AI feedback, and post-task trust measures using jsPsych

Participants:

- Planned for human participants; final analyses conducted using simulated data to validate study design and analysis pipeline

Timeline:

- Study design, IRB preparation, implementation, and analysis completed over approximately 3 months

Challenge:

- Designing AI errors that were realistic and interpretable without revealing experimental intent
- Balancing experimental control with an interaction flow that felt intuitive to participants

Design & Implementation Takeaways

Experiment Design:

- Designed a controlled human-AI interaction paradigm that isolates AI error type (false alarms vs. misses) as a UX-relevant variable
- Built a fully interactive web-based experiment in jsPsych, including scenario onboarding, knowledge checks, trial logic, AI advice behavior, and surveys

Possible UX Impact:

- Demonstrates that different AI error types (false alarms vs. misses) produce distinct effects on user trust, reliance, and correction behavior
- Indicates that AI systems should tailor feedback and uncertainty communication based on error type to support appropriate reliance
- Highlights the risk of trust miscalibration when AI errors are frequent but poorly contextualized within the interaction flow

Reflection:

- Demonstrated the ability to translate abstract human-factors questions into a deployable research instrument
- Reinforced the importance of testing interaction logic early to identify UX risks before running live studies.

Project 3: Behavioral and fMRI Analysis of Trust and Reciprocity

Analyzing behavioral and fMRI data from a trust game to examine how social decision-making and reciprocity vary across conditions and individuals. Leading data preprocessing and statistical analysis to translate experimental data into interpretable behavioral and neural measures.

Background

Current Landscape

- Trust decisions are shaped by social uncertainty and asymmetric incentives
- Framing effects can shift how people interpret identical outcomes, changing willingness to take social risk

Knowledge Gap

- We still don't fully understand when framing changes trust behavior, especially under high-risk payoff structures where betrayal is most costly
- Less is known about how framing and payoff features relate to expectations of reciprocity and neural responses during trust decisions

Future Relevance

- Clarifying how framing and payoff context shape trust can inform how we design high-stakes choices (e.g., financial, health, platform safety) where users must rely on others
- Identifying behavioral/neural markers of risk-sensitive trust can help predict when trust breaks down—and what conditions support stable cooperation.

Research Question: How do gain vs. loss framing and trial-level risk/temptation influence (1) trust decisions, (2) predictions of reciprocity, and (3) associated brain activity?

Specs

Role:

- Led behavioral and fMRI data analysis, including preprocessing, modeling, and interpretation of trust and reciprocity measures
- Developed the full analysis plan and analytical framework for the study

Methods:

- Behavioral analysis of trust decisions and reciprocity predictions across framing and payoff conditions using R
- fMRI preprocessing and first-level modeling using SPM12, MATLAB, and Python with planned group-level analyses linking neural activity to decision variables

Participants:

- Existing dataset collected in a prior fMRI trust game study; adult participants scanned across two task runs with multiple trial types

Timeline:

- Ongoing analysis conducted over the current academic year

Challenge:

- Designing an analysis pipeline that integrates behavioral and neural data while maintaining interpretability
- Managing analytic complexity across multiple conditions, runs, and outcome measures

Analysis Focus & Early Takeaways

Analysis Focus:

- Quantifying how gain vs. loss framing and payoff structure shape trust decisions and expectations of reciprocity
- Linking trial-level decision variables to neural responses associated with risk evaluation and social decision-making

Future UX Impact:

- Shows that framing and payoff structure meaningfully shape trust decisions and expectations of reciprocity, even when outcomes are identical
- Highlights trust as a dynamic, risk-sensitive process rather than a stable user trait, with implications for system design under uncertainty
- Informs the design of high-stakes systems (e.g., financial, health, safety) where framing can unintentionally amplify perceived social risk

Reflection:

- Reinforced the value of integrating behavioral and neural data to interpret trust-related decisions more precisely
- Strengthened skills in designing interpretable analysis pipelines for complex, multi-condition datasets

Additional Projects and Experience

Concussion Tracking App (Hackathon)

- Collaborated on a rapid prototype for at-home concussion monitoring; led decisions around cognitive testing and accessibility considerations (e.g., reduced visual strain)
- Won best project in a 48-hour hackathon, emphasizing feasibility under extreme time constraints

Participant Recruitment & Study Support

- Supported participant recruitment and coordination for an ongoing behavioral research study, ensuring protocol adherence and clear participant communication
- Gained hands-on experience with the operational side of research execution

Audio Processing Research Lab

- Assisted with participant sessions for a psychology study on auditory processing, including task administration and data collection
- Built early experience working within structured experimental protocols and research teams